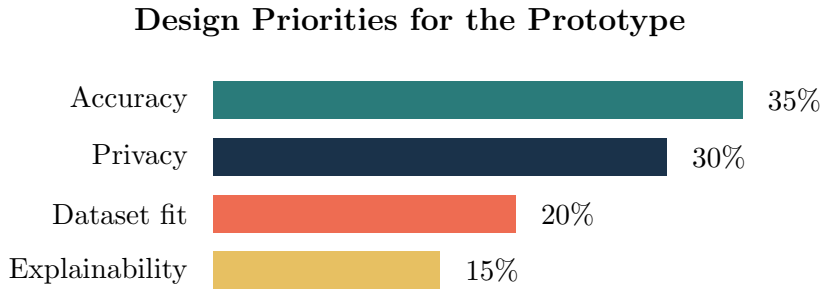
Software Design Document

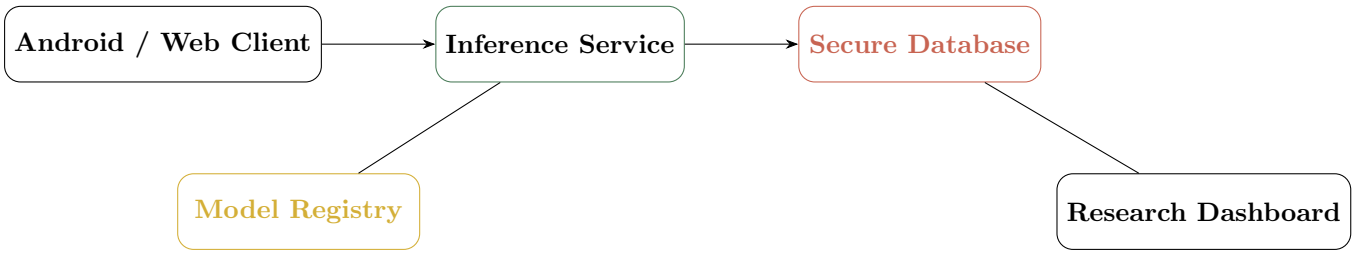
Architecture for Facial Expression-Based Depression Assessment Using Mobile Vision

1. Design Overview

The design follows a modular pipeline: consent and input capture, quality checks, face preprocessing, feature extraction, temporal modeling, evaluation and result reporting. The implementation can be built as a Python research prototype with a Streamlit/FastAPI interface, while the mobile behavior is simulated using uploaded clips or webcam capture.



2. Architecture

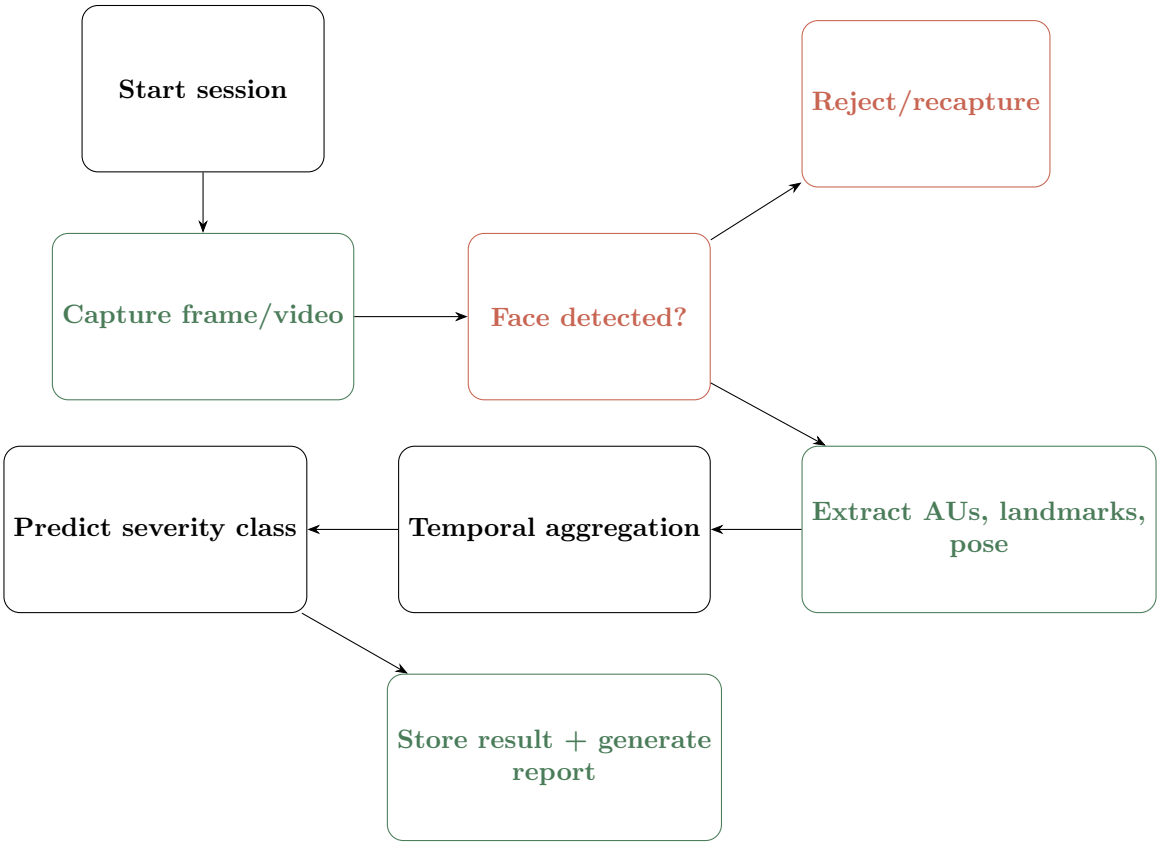


For coursework, the client can be a web page or notebook upload form. The inference service handles OpenCV/MediaPipe/Open Face feature extraction and PyTorch/scikit-learn prediction. The database can be a local SQLite file or CSV feature store, because the project should remain simple and reproducible.

3. Component Design

Component	Responsibility
Capture Module	Loads benchmark video, image batches or webcam frames; records session metadata and consent flag.
Quality Gate	Detects face presence, blur, brightness and frame validity before feature extraction.
Preprocessor	Normalizes frame size, crops face, aligns landmarks and samples fixed-length frame windows.
Feature Extractor	Produces landmarks, action units, pose, gaze and optional expression probabilities.
Temporal Model	Uses GRU/LSTM/RNN or simpler Random Forest baseline on aggregated features.
Evaluator	Computes classification and regression metrics, confusion matrix and error plots.
Report Generator	Exports subject/session result, disclaimer, metrics and cited references.

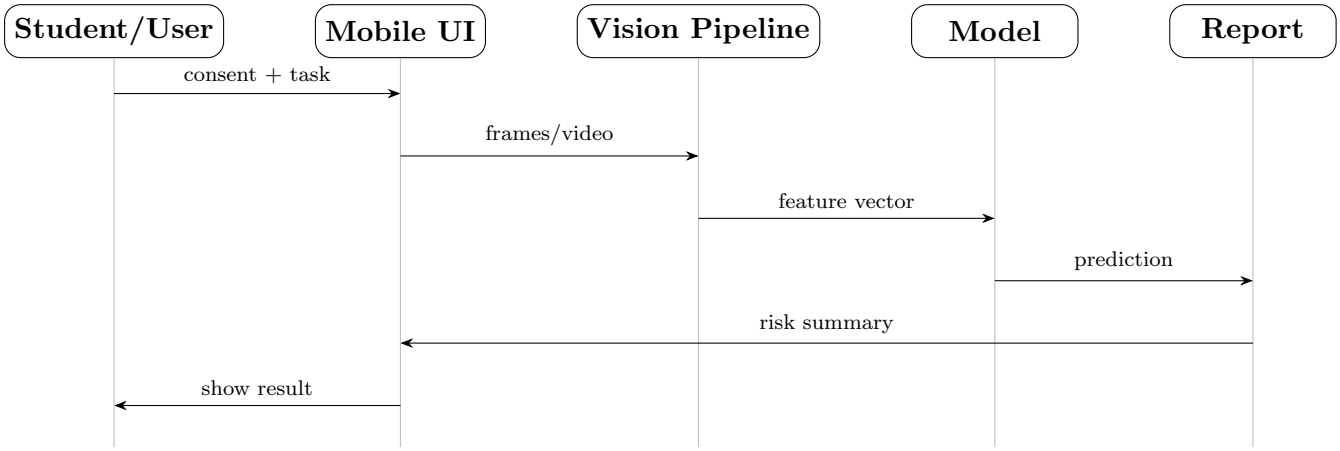
4. Data Flow



5. Detailed Processing Pipeline

Stage	Input	Processing	Output
Ingest	Video/image	Read frames, assign session ID, sample frames at fixed interval	Frame sequence
Quality	Frames	Face/no-face check, blur threshold, brightness threshold	Valid frames
Face analysis	Valid frames	Crop, align, extract 2D/3D landmarks, AUs and pose	Feature matrix
Temporal encoding	Feature matrix	Sliding windows, normalization, RNN/GRU or aggregate statistics	Embedding/score
Prediction	Embedding	Classify risk or regress score	Predicted label/severity
Evaluation	Predictions + labels	Metrics and plots	Results table

6. Sequence Diagram



7. Model Choices

Recommended baseline stack: MediaPipe Face Mesh or OpenFace for landmarks/AUs, scikit-learn Random Forest for a transparent first baseline, and PyTorch GRU/LSTM for the FacialPulse-inspired temporal model. If time is short, implement Random Forest plus GRU and compare them on the same extracted feature files.

Model	Why	Expected Output
Random Forest	Explainable, strong in Mood-Capture for 3D landmarks	Class label and feature importance
GRU/LSTM	Captures temporal facial dynamics like FacialPulse	Severity score or class probability
CNN/ResNet-50	Frame-level visual baseline from alignment/scheduling papers	Frame embedding or prediction
CCANet/FPN-inspired	Strong AVEC 2014 reference	Advanced stretch goal

8. Database And File Design

Entity/File	Fields	Purpose
`sessions.csv`	session_id, dataset, subject_id_hash, consent, timestamp	Track experiment sessions
`frames/`	session_id/frame_id.jpg or extracted sample	Optional local debugging only
`features.parquet`	session_id, frame_id, AU*, landmark*, pose*, gaze	Reusable model input
`labels.csv`	session_id, PHQ/BDI score, binary label, split	Ground truth for training/testing
`metrics.json`	model, split, accuracy, F1, MAE, RMSE	Evaluation record
`reports/`	session report PDF/HTML	Human-readable output

9. UI Design

- **Home/Test screen:** upload video or select sample dataset clip, then run analysis.
- **Results screen:** risk label, score, confidence, quality warnings and short explanation.
- **Evaluation screen:** dataset statistics, confusion matrix, MAE/RMSE chart and model comparison.
- **References/About screen:** dataset URL, papers used, ethical disclaimer and limitations.

10. Security, Privacy And Ethics Design

The system should minimize raw-image storage, anonymize IDs, separate labels from images, and keep any facial data local. The UI must state that outputs are experimental and should not be used for diagnosis, grading, employment, insurance or disciplinary decisions. Any real participant data requires informed consent and institutional approval.

11. Testing Plan

Test Type	Test Cases	Pass Criteria
Unit	Frame reader, face detector, feature extractor, metric functions	All functions return expected schema on sample files
Integration	Run full pipeline on sample clip	Feature file, prediction and report are generated
Model	Train/test split evaluation	Metrics reproducible with fixed seed
Robustness	No face, dark frame, blurred frame	System gives quality warning instead of crashing
Usability	Student demo walkthrough	A user can run a sample analysis in under 3 minutes

12. Risks And Mitigations

Risk	Mitigation
Dataset access delay	Use AVEC 2014/DAIC-WOZ if available; otherwise demonstrate on public sample clips and report literature metrics separately.
Low model accuracy	Frame as research prototype; compare simple baselines and explain limitations.
Privacy concerns	Avoid storing raw face images; use feature-only files for demo.
Bias across gender/race/lighting	Report subgroup limitations and do not claim clinical validity.
Medical misuse	Use clear disclaimer and restrict output to screening/research language.

13. References

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